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EXPERIMENT 1

WINDOWS FUNDAMENTALS

AIM:

To study the Windows operating system architecture, system configuration tools, file systems, processes, registry, and built-in security mechanisms.

ALGORITHM:

1. Launch and access a Windows operating system.
2. Explore system components, folders and configuration tools.
3. Analyse processes, services and system management utilities.
4. Study Windows security features and permissions.
5. Observe monitoring, protection and auditing mechanisms.

TASKS:

Task 1 Windows Editions

The Windows operating system has a long history dating back to 1985, and currently, it is the dominant operating system in both home use and corporate networks. Because of this, Windows has always been targeted by hackers & malware writers.

Windows XP was a popular version of Windows and had a long-running. Microsoft announced Windows Vista , which was a complete overhaul of the Windows operating system. There were many issues with Windows Vista. It wasn't received well by Windows users, and it was quickly phased out.

When Microsoft announced the end-of-life date for Windows XP, many customers panicked. Corporations, hospitals, etc., scrambled and tested the next viable Windows version , which was Windows 7, against many other hardware and devices. Vendors had to work against the clock to ensure their products worked with Windows 7 for their customers. If they couldn't, their customers had to break their agreement and find another vendor that upgraded their products to work with Windows 7. It was a nightmare for many, and Microsoft took note of it.

Windows 7, as quickly as it was released soon after, was marked with an end of support date. Windows 8.x came and left and it was short-lived, like Vista.

Then arrived Windows 10 followed by Windows 11, which is the current Windows operating system version for desktop computers.

Windows 11 comes in 2 flavors, Home and Pro. You can read the difference between the Home and Pro here.

Even though we didn't talk about servers, the current version of the Windows operating system for servers is Windows Server 2025.

Many critics like to bash on Microsoft, but they have made long strides to improve the usability and security with each new version of Windows .

Note : The Windows edition for the attached VM is Windows Server 2019 Standard, as seen in **System Information** .

Update : As of June 2021, Microsoft announced the retirement dates for Windows 10 here .

" *Microsoft will continue to support at least one Windows 10 Semi-Annual Channel until October 14, 2025* ".

As of October 5th, 2021 - Windows 11 now is the current Windows operating system for end-users. Read more about Windows 11 here .

Answer the questions below

What encryption can you enable on Pro that you can't enable in Home?

BitLocker ✓ Correct Answer

Task 2 The Desktop (GUI)

The Windows Desktop, aka the graphical user interface or GUI in short, is the screen that welcomes you once you log into a Windows 10 machine.

Traditionally, you need to pass the login screen first. The login screen is where you need to enter valid account credentials; usually, a username & password of a preexisting Windows account on that particular system or in the Active Directory environment (if it's a domain-joined machine).

Answer the questions below

Which selection will hide/disable the Search box?

Hidden — ✓ Correct Answer

Which selection will hide/disable the Task View button?

Show Task View button — ✓ Correct Answer

Besides Clock and Network, what other icon is visible in the Notification Area?

Action Center — ✓ Correct Answer

Task 3 Introduction to Windows

▶ Start Machine

The Windows operating system (OS) is a complex product with many system files, utilities, settings, features, etc.

This module will attempt to provide a general overview of just a handful of what makes up the Windows OS, navigate the user interface, make changes to the system, etc. The content is aimed at those who wish to understand and use the Windows OS on a more comfortable level.

Press the **Start Machine** button below to launch the attached virtual machine.

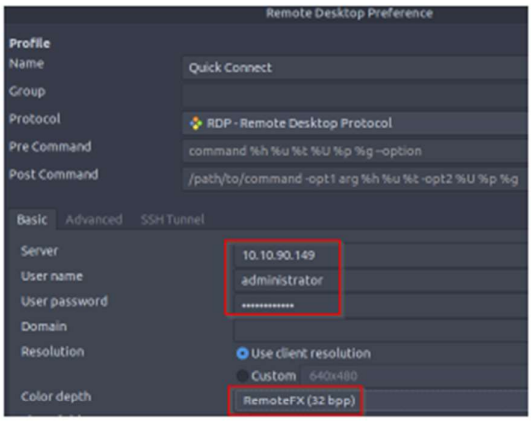
The virtual machine should open within your web browser.

If you want to access the virtual machine via Remote Desktop , use the credentials below.

Machine IP : `MACHINE_IP`

User : `administrator`

Password : `letmein123!`



Accept the Certificate when prompted, and you should be logged into the remote system now.

Note : The virtual machine may take up to 3 minutes to load.

Answer the questions below

Read above and start the virtual machine.

No answer needed — ✓ Correct Answer

Task 4 The File System

The file system used in modern versions of Windows is the **New Technology File System** or simply NTFS .

Before NTFS, there was **FAT16/FAT32** (File Allocation Table) and **HPFS** (High Performance File System).

You still see FAT partitions in use today. For example, you typically see FAT partitions in USB devices, MicroSD cards, etc. but traditionally not on personal Windows computers/laptops or Windows servers.

NTFS is known as a journaling file system. In case of a failure, the file system can automatically repair the folders/files on disk using information stored in a log file. This function is not possible with FAT.

NTFS addresses many of the limitations of the previous file systems; such as:

- Supports files larger than 4GB
- Set specific permissions on folders and files
- Folder and file compression
- Encryption (Encryption File System or **EFS**)

If you're running Windows, what is the file system your Windows installation is using? You can check the Properties (right-click) of the drive your operating system is installed on, typically the C drive (C:\).

Answer the questions below

What is the meaning of NTFS?

New Technology File System — ✓ Correct Answer

Task 5 The Windows\System32 Folders

The Windows folder (`C:\Windows`) is traditionally known as the folder which contains the Windows operating system.

The folder doesn't have to reside in the C drive necessarily. It can reside in any other drive and technically can reside in a different folder.

This is where environment variables, more specifically system environment variables, come into play. Even though not discussed yet, the system environment variable for the Windows directory is `%windir%` .

Per Microsoft , " *Environment variables store information about the operating system environment. This information includes details such as the operating system path, the number of processors used by the operating system, and the location of temporary folders* ".

Answer the questions below

What is the system variable for the Windows folder?

%windir% — ✓ Correct Answer

Task 6 User Accounts, Profiles, and Permissions

User accounts can be one of two types on a typical local Windows system: **Administrator** & **Standard User**.

The user account type will determine what actions the user can perform on that specific Windows system.

- An Administrator can make changes to the system: add users, delete users, modify groups, modify settings on the system, etc.
- A Standard User can only make changes to folders/files attributed to the user & can't perform system-level changes, such as install programs.

You are currently logged in as an Administrator. There are several ways to determine which user accounts exist on the system.

One way is to click the `Start Menu` and type `Other User`. A shortcut to `System Settings > Other users` should appear.

Answer the questions below

What is the name of the other user account?

tryhackmebilly — ✓ Correct Answer

What groups is this user a member of?

Remote Desktop Users,Users — ✓ Correct Answer

What built-in account is for guest access to the computer?

Guest — ✓ Correct Answer

What is the account description?

window$Fun1! — ✓ Correct Answer

Task 7 User Account Control

The large majority of home users are logged into their Windows systems as local administrators. Remember from the previous task that any user with administrator as the account type can make changes to the system.

A user doesn't need to run with high (elevated) privileges on the system to run tasks that don't require such privileges, such as surfing the Internet, working on a Word document, etc. This elevated privilege increases the risk of system compromise because it makes it easier for malware to infect the system. Consequently, since the user account can make changes to the system, the malware would run in the context of the logged-in user.

To protect the local user with such privileges, Microsoft introduced **User Account Control** (UAC). This concept was first introduced with the short-lived Windows Vista and continued with versions of Windows that followed.

Note : UAC (by default) doesn't apply for the built-in local administrator account.

How does UAC work? When a user with an account type of administrator logs into a system, the current session doesn't run with elevated permissions. When an operation requiring higher-level privileges needs to execute, the user will be prompted to confirm if they permit the operation to run.

Answer the questions below

What does UAC mean?

User Account Control ✓ Correct Answer

Task 8 Settings and the Control Panel

On a Windows system, the primary locations to make changes are the Settings menu and the Control Panel.

For a long time, the Control Panel has been the go-to location to make system changes, such as adding a printer, uninstall a program, etc.

The Settings menu was introduced in Windows 8, the first Windows operating system catered to touch screen tablets, and is still available in Windows 10. As a matter of fact, the Settings menu is now the primary location a user goes to if they are looking to change the system.

Answer the questions below

In the Control Panel, change the view to **Small icons**. What is the last setting in the Control Panel view?

Windows Defender Firewall ✓ Correct Answer

Task 9 Task Manager

The last subject that will be touched on in this module is the **Task Manager**.

The Task Manager provides information about the applications and processes currently running on the system. Other information is also available, such as how much CPU and RAM are being utilized, which falls under **Performance**.

Answer the questions below

What is the keyboard shortcut to open Task Manager?

Ctrl+Shift+Esc ✓ Correct Answer

Task 10 Conclusion

Again, this was a generic overview of the Windows OS.

There are intermediate and advanced topics for each topic (task) that was covered in this room.

Hence, **Task 9** ended with a detailed blog post explaining the Task Manager in great detail.

In future modules, we'll cover topics like the Windows folder, the management console, security tools (Windows Defender, Windows Firewall, etc.), to name a few.

Answer the questions below

Read above and terminate the Windows machine you deployed in this room.

No answer needed ✓ Correct Answer

RESULT:

All the tasks in windows fundamentals tryhackme room were completed successfully.